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1: Library management -system - Identify candidate classes (Books, Readers, Loans, Employees, Publishers) + Books: main entity, book information management.

- + DocGia: person who borrows books.
- + PhieuMuon: records borrowing/returning.
- + NhanVien: system manager (librarian, warehouse keeper...).
- + NhaXuatBan: book source information.

- Properties and behavior (methods) of each class + Book

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- Attributes: maSach (PK, string), nameSach (string), tacGia (string), namXuatBan (int), maNXB (FK), soLuongTon (int), thai page (string: "Available"/"On loan").
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- Behavior: addSach(), capNhatSach(), deleteSach(), saveSach(), capNhatTonKho().

+ DocGia

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- Attributes: maDocGia (PK, string), hoTen (string), daySinh (date), diaChi (string), soDienThoai (string), dayDangKy (date), gioiTinh (boolean).
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- Behavior: dangKyThe (), capNhatThongTin (), seeLichSuMuon (). + PhieuMuon
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- Attributes: maPhieu (PK, string), maDocGia (FK), maSach (FK), maNhanVien (FK), nhaMuon (date), nayHenTra (date), ngoTraThucTe (date), tinhTrang (string: "On loan"/"Paid"/"Overdue").
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- Behavior: createPhieuMuon(), searchSach(), saveTraQuaHan(), crystalPhiPhat(). + NhanVien

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- Attributes: maNhanVien (PK, string), hoTen (string), chucVu (string), date of birth (date), soDienThoai (string).
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- Behavior: dangNhap (), quanLyDocGia (), quanLyMuonTra (), quanLySach ().()

+ NhaXuatBan

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- Attributes: maNXB (PK, string), nameNXB (string), diaChi (string), email (string).

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- Behavior: addNXB (), capNhatNXB ().

- Relationships

□ Books and Publishers: 1 book – 1 publisher

- ☐
- ☐ Readers and Vouchers: 1 reader – many borrowing slips
- ☐
- ☐ Books and coupons: 1 book - many coupons
- ☐
- ☐ Employees and Vouchers: 1 employee – many vouchers

LESSON 2: Online sales management system

- Identify main use cases

Main actors:

- + Customer (Customers / Potential Customers).
- + NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

- ☐
- ☐ DangKyTaiKhan (Customer registration).
- ☐
- ☐ DangNhap (Sign in).
- ☐
- ☐ See SanPham (Search / View product list).
- ☐
- ☐ Add to Cart (Add to Cart).
- ☐
- ☐ DatHang (Order).
- ☐
- ☐ ThanhToan (Online payment / COD).
- ☐
- ☐ View DonHang (View order history).

Admin / NguoiQuanLy:

- ☐
- ☐ QuanLySanPham (Add/Edit/Delete products).
- ☐
- ☐ QuanLyDonHang (View/Confirm/Cancel order).
- ☐
- ☐ QuanLyKhachHang (Customer information management).
- ☐

- ☐ QuanLyThanhToan (Payment processing, refund if necessary).

Extend / Include:

- ☐
- ☐ ThanhToan includes XacNhanDonHang.
- ☐
- ☐ DatHang extends ThanhToan (if paying online).

- From use case, find the necessary classes

- ☐
- ☐ Customers (Customer): from register, view orders.
- ☐
- ☐ SanPham (Product): product management.
- ☐
- ☐ GioHang (Cart): temporarily save the product before ordering.
- ☐
- ☐ Donate (Order): from order, payment.
- ☐
- ☐ ChiTietDonHang (OrderDetail): product details in the order.
- ☐
- ☐ ThanhToan (Payment): payment processing (may have subclasses such as OnlinePayment, COD).
- ☐
- ☐ Admin or Personnel (inherited from User if there is a user system).

- Analyze relationships between classes (specify inheritance if any)

☐ KhachHang and DonHang: 1 customer – many orders

☐ DonHang and ChiTietDonHang: 1 order – many order details

(Composition)

☐ ChiTietDonHang and SanPham: many single details – 1 product

☐ DonHang and ThanhToan: 1 order – 1 payment

☐ User (abstract class) ← extends ← Customer and Admin (Customer and Admin are both the same type of User, sharing the same account, password, and full name).

Lesson 3: Convert class diagrams into data tables

1. MonHoc table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maMon	Nvarchar(10)	x		x	x
2	tenMon	Nvarchar(50)			x	
3	soTinChiLT	Int			x	
4	soTinChiTH	Int			x	

2. Science Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maKhoaHoc	Nvarchar(10)	x			x
2	tenKhoaHoc	Nvarchar(50)			x	
3	ngayTao	Date			x	